Andrei Iuliu Ghita

andrei.ghita2002@gmail.com
linkedin.com/in/andrei-iuliu-ghita-dev/ | github.com/AndreiGhita2002

EDUCATION

University of Glasgow | Glasgow, UK **Jun 2026**
Integrated Master in Computing Science, MSci
First Class Honours (1:1)

International School Basel | Basel, Switzerland **May 2021**
International Baccalaureate Diploma Programme
38/45 points achieved – achieved 6,6,6,7 in HL Subjects (equivalent to A*AAA)

PROFESSIONAL EXPERIENCE

Research Internship | University of Glasgow | Glasgow, UK **Summer 2026**
- Worked with the Glasgow Interactive SysTems (GIST) group on mixed reality spatial-computing application, enabling users to locate real-world objects through passthrough AR and on-device guidance.
- My work will contribute to a future research paper.

Graduate Teaching Assistant | University of Glasgow | Glasgow, UK **Sep 2025 - April 2026**
- Instructing students in fundamental data science, translating complex topics into digestible tutorials.
- Cultivating a supportive and open learning environment, earning student trust as a reliable resource for learning.
- Developing strong communication, mentorship, and problem-solving skills by providing tailored, constructive feedback.

ACADEMIC PROJECTS

Dynamic Pipeline AI Model Inference – Masters Dissertation **Sep 2025 - May 2026**
- Developed a system that dynamically optimises a inference pipeline by rebalancing stages at runtime based on real-time profiler data and system load. Achieved 30% performance improvements on busy clusters.
- Built on top of PyTorch; supports most AI model architectures on both CPU and CUDA devices.
- Paper will be presented and published in the Euro-Par conference.

GPU-based Virtual Machine – Bachelor Dissertation **Sep 2024 - May 2025**
- Designed and implemented a novel Virtual Machine executing directly on GPU kernels.
- Engineered the VM to simultaneously run both compute workloads and real-time graphical programs.
- Developed the CPU-side host application in C++ with Vulkan to manage resource and program uploads to the GPU.
- Ensured high portability by compiling the GPU-side VM (written in GLSL) to the SPIR-V intermediate.

Team Project – Research website for School of Psychology **Sep 2023 - May 2024**
- Delivered a Django/React-based games platform for the School of Psychology, designed to meet client requirements.
- Played a key role in refactoring the application into a decoupled stack, implementing a React frontend with a Django REST API.
- Architected the PostgreSQL database schema, modelling data relationships to ensure integrity and optimise queries.
- Deployed the full-stack application using Docker for containerisation and Nginx as a reverse proxy.

EXTRA-CURRICULAR EXPERIENCE

Hackathon Win | "Do you have the GUTS?" Hackathon **October 2025**
- Secured 1st place in the 24-hour Free to Book challenge by developing a web game for their 404 error page.
- Delivered a polished web game by prioritising UI design and brand alignment with the client's existing website.

Tech Society Committee Member | GUTS | Glasgow, UK **Sep 2025 - May 2026**
- Organising and planning society events, with 200+ attendees, including social events and hackathons.

SKILLS

Systems & Performance: C/C++, Rust, Go, CMake, multithreading (pthreads, OpenMP, Tokio)
Machine Learning: PyTorch, NumPy, LangSmith, Ollama
Web & Backend: Django, Flask, React, PostgreSQL, MongoDB, Docker, Nginx
GPU & Graphics: Vulkan, OpenGL, CUDA, GLSL, SPIR-V, Unity
Programming Languages: C/C++, Rust, Go, Python, Java, C#, Haskell, JavaScript/TypeScript
Other Classes: Digital and Embedded Electronics, Linear Algebra, Physics
Languages: English - Fluent, German - Intermediate, Romanian - Native